

Work done (W) = Force (F) x displacement (s)

$$W = Fs$$

Unit: **Nm** or **J**

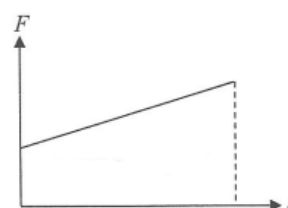
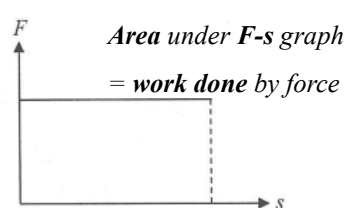
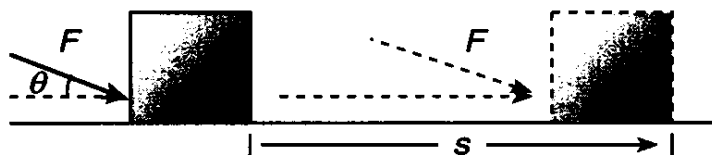
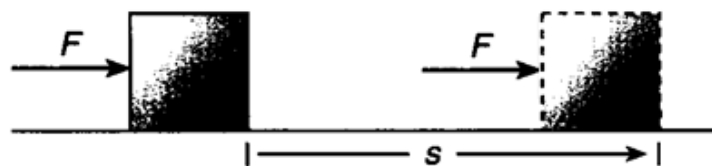
Remarks:

- Work is a process to transfer energy by the application of force through a displacement.
- Work is a **scalar**.
- If the force and displacement are **not in same direction**, then only the component of force **parallel to** the displacement will do work on object.

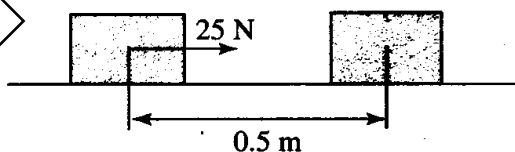
$$W = Fscos\theta$$

(θ = angle between F and s)

- If the applied force is **perpendicular** to the displacement, **no work is done** by the force.

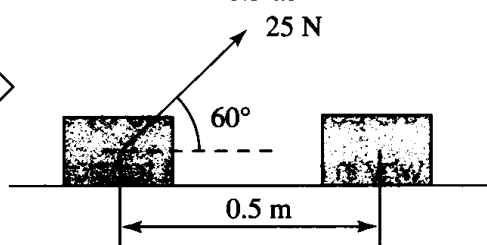


Motion



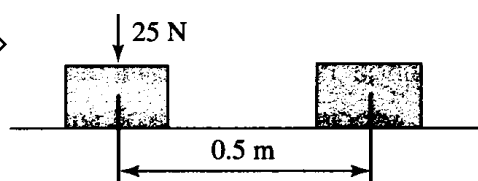
$$W = 25 \times 0.5 = 12.5 \text{ Nm}$$

Motion



$$W = 25 \times 0.5 \times \cos 60^\circ = 6.25 \text{ Nm}$$

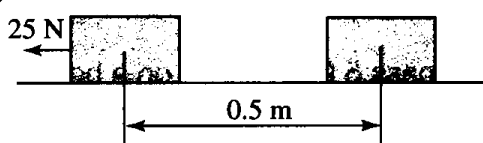
Motion



$$W = 0 \text{ Nm}$$

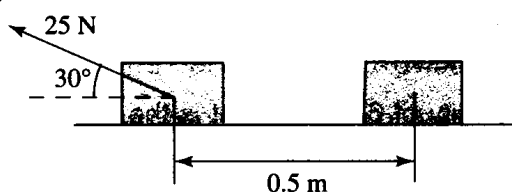
No work is done by the force

Motion



$$W = -25 \times 0.5 = -12.5 \text{ Nm}$$

Motion



$$W = -25 \times 0.5 \times \cos 30^\circ = -10.8 \text{ Nm}$$

Kinetic energy

- A **moving object** possesses kinetic energy (KE). The faster it moves, the more KE it possesses.
- Consider a constant force F acting on an object, due to the net force, the object accelerates uniformly from rest to velocity v after travelling for a displacement s :

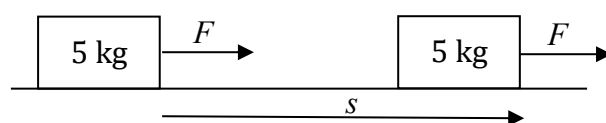
$$W = Fs = (ma)\left(\frac{v^2}{2a}\right) = \frac{1}{2}mv^2$$

∴ **Kinetic energy** of an object of mass m at a velocity v :

$$KE = \frac{1}{2}mv^2$$

At rest

$v = 10 \text{ ms}^{-1}$



KE gain = 250 J

Work done = 250 J

Remarks:

Energy is scalar and its unit is Joule (J)

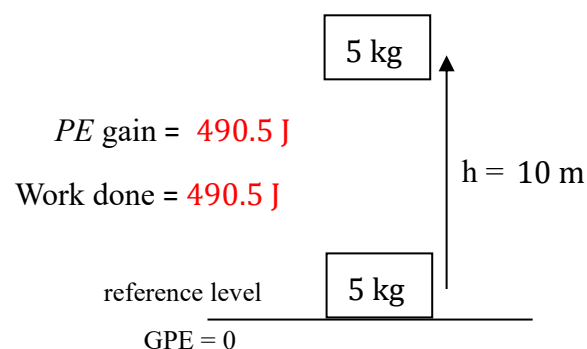
Gravitational potential energy

- Gravitational potential energy (GPE)** is the energy possessed by an object due to its position under gravity. When an object is raised to higher position, it gains GPE.
- Consider an object of mass m is lifted by a force F (equals to mg) to a height h above the ground at a constant velocity.

$$W = Fs = (mg)(h)$$

∴ **Potential energy** of an object at height h :

$$PE = mgh$$



PE gain = 490.5 J

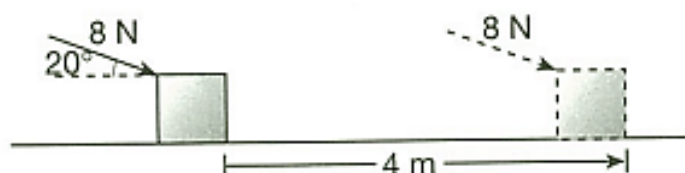
Work done = 490.5 J

- A 3 kg block is pushed from rest along a smooth surface for 4 m by a 8 N force which makes an angle of 20° with the horizontal.

- Find the work done by the force
- What is the final velocity of the block?

$$\begin{aligned} \text{a. } WD &= 8 \cos 20^\circ (4) \\ &= 30.1 \text{ J} \end{aligned}$$

$$\text{b. } K.E_{\text{gain}} = 30.1 \text{ J} = \frac{1}{2}(3)v^2 \quad v = 4.48 \text{ ms}^{-1}$$



- A car of mass 1045 kg is travelling at 30 km h^{-1} . When the brakes are applied, the total friction on the tyres of the car is 1200 N.

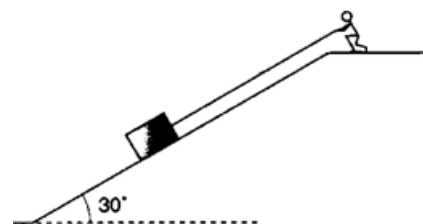
- What is the KE of the car when it is travelling at 30 km h^{-1} ?
- The brakes are applied until the car stops. How long does the car travel before it stops?

$$\text{a. } KE = \frac{1}{2}(1045)(8.33)^2 = 36285 \text{ J}$$

$$\begin{aligned} \text{b. } KE_{\text{loss}} &= WD \\ 36285 &= 1200d \quad d = 30.2 \text{ m} \end{aligned}$$

3. A man uses a wire to trail a heavy box with a constant speed on a slope to a height of 5 m. The mass of the box is 30 kg and the friction between the box and the slope is 20 N.

- Find the increase in potential energy of the box.
- Find the work done against friction.
- Find the tension in the wire.
- What is the work done by the normal reaction acting by the slope? Explain briefly.



a. $PE_{\text{gain}} = 5(30)(9.81) = 1471.5J$

b. $WD = 20 \left(\frac{5}{\sin 30^\circ} \right) = 200J$

c. $T = 30(9.81)(\sin 30^\circ) + 20 = 167.15N$

d. Work done = 0

The normal reaction is perpendicular to the direction of motion of the box

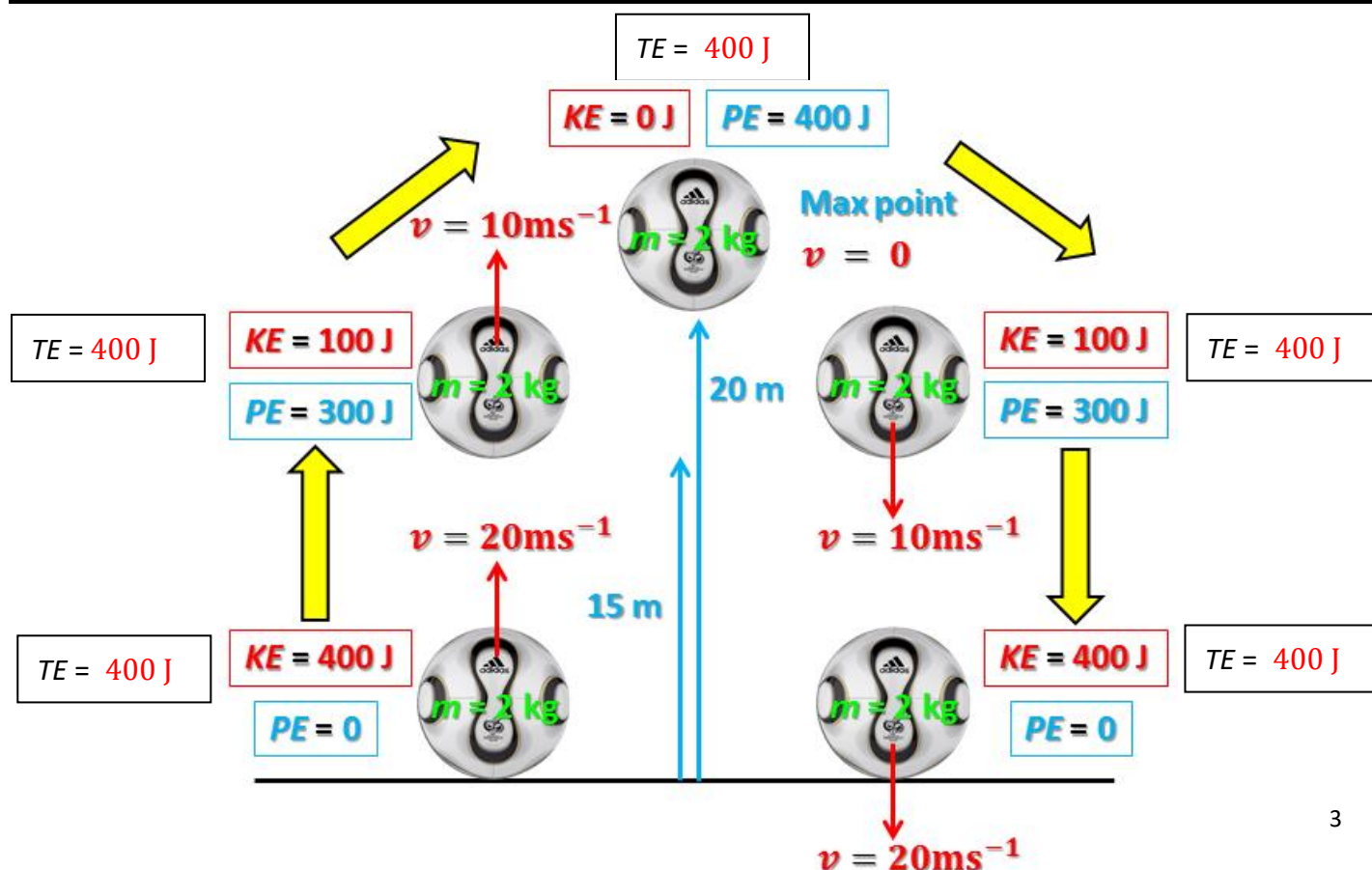
Law of conservation of energy

Energy cannot be created or destroyed but can only be changed from one form to another

Total mechanical energy (TE) = Kinetic energy (KE) + Potential energy (PE)

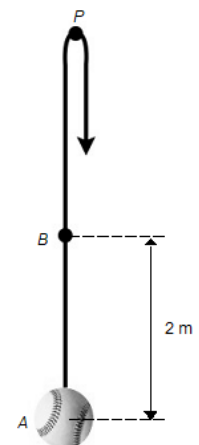
Remarks:

- If air resistance is negligible, there is no work done on the body during the flight in air. Thus, the total mechanical energy of the body is **conserved**.
- During the **upward** motion: K.E $\rightarrow$ P.E
- During the **downward** motion: P.E $\rightarrow$ K.E



4. A baseball of mass 145 g is projected vertically upwards with a velocity of 10 m s^{-1} from position A . Neglect air resistance and take the potential energy at A as zero.
- (a) Find the
- potential energy, and
 - kinetic energy of the ball at position B , which is 2 m away from A .
- (b) A student suggests that the mechanical energy of the ball at B is equal to that at P . Is he correct? Explain briefly.

- (a) (i) PE of the ball at $B = mgh = 0.145 \times 9.81 \times 2 = 2.8449 \approx 2.85 \text{ J}$ (1M+1A)
 (ii) Applying $v^2 - u^2 = 2as$, we have
 $v^2 - 10^2 = 2 \times (-9.81) \times 2$
 $v^2 = 60.76$ (1M)
 KE of the ball at $B = \frac{1}{2}mv^2 = \frac{1}{2} \times 0.145 \times 60.76 = 4.4051 \approx 4.41 \text{ J}$. (1A)
 (b) The student is correct (1A). Since no energy is lost in overcoming air resistance, the mechanical possessed by the ball is constant (1A).



5. A pendulum bob of mass 0.2 kg is suspended by a light inextensible string. The bob is projected at P with a speed of 0.8 m s^{-1} as shown. It reaches the highest position S on the first swing.
- (a) Describe the energy change as the bob moves from P to S .
- (b) What is the speed of the bob at the lowest position Q ?
- (c) What is the vertical distance between S and Q ?
- (d) A student suggests that the tension in the string does work on the bob. Is he correct? Explain briefly.
- (e) Suppose the string breaks when the bob moves to R .
- (i) Draw a free body diagram to show all the forces acting on the bob just before the string breaks.

- (a) As the bob moves from P to Q , potential energy is converted into kinetic energy. (1A) As the bob moves from Q to S , kinetic energy is converted into potential energy (1A).

- (b) Take PE at Q as zero. By conservation of energy,
 $KE_P + PE_P = KE_Q + PE_Q$
 $\frac{1}{2} \times 0.2 \times 0.8^2 + 0.2 \times 9.81 \times 0.3 = \frac{1}{2} \times 0.2 \times v_Q^2 + 0$
 $v_Q = 2.5546 \text{ m s}^{-1}$ (1M+1A)

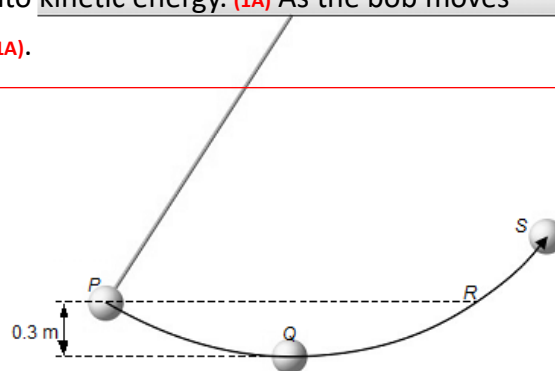
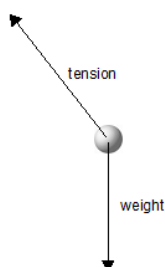
The speed of the bob at Q is 2.55 m s^{-1} .

- (c) By conservation of energy,
 $KE_P + PE_P = KE_Q + PE_Q$
 $\frac{1}{2} \times 0.2 \times 2.5546^2 + 0 = 0 + 0.2 \times 9.81 \times h$
 $h = 0.3326 \text{ m}$ (1M+1A)

The vertical distance between S and Q is 0.333 m.

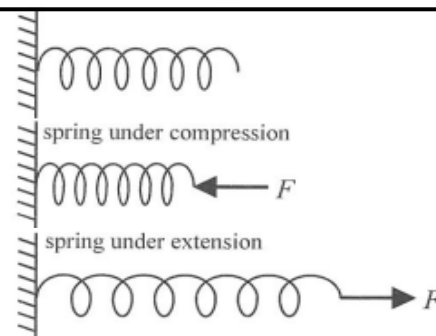
- (d) The student is not correct (1A). As the tension in the string is always perpendicular to bob's direction of motion, no work is done by the tension (1A).

- (e) (i)



Elastic potential energy

- **Elastic potential energy (EPE)** is the energy possessed by an elastic object when the object is stretched, compressed or bent.
- When the deformed object restored to its original shape, the elastic potential energy is released.



6. Mark goes bungee jumping. He falls from rest from point O which is 40 m above a river. When he reaches point A which is 15 m below O , the elastic string starts to stretch. When he reaches the lowest point B , the elastic string is elongated by 10 m. The mass of Mark is 50 kg. Neglect the mass of the string and air resistance. Take the gravitational potential energy at the surface of the river as zero.

- (a) State the energy transformation process when Mark moves from O to B . (2 marks)

(a) When Mark moves from O to A , his gravitational potential energy is transformed into kinetic energy.

When he moves from A to B , his kinetic energy and gravitational potential energy are transformed into elastic potential energy.



7. The above diagram shows the track of a roller coaster in an amusement park. The coaster of mass 200 kg is released from point A . By the action of gravity, it goes through B , C and finally reaches D with a speed of 20 ms^{-1} . Then it is braked uniformly to stop at a point E , which is 15 m away from D .

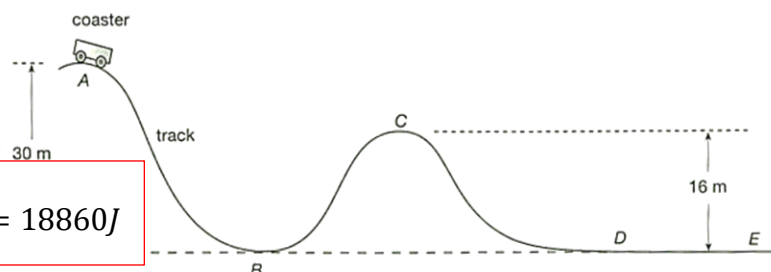
- (a) Find the loss in potential energy of the coaster in its journey from A to D . (Take $g = 9.81 \text{ ms}^{-2}$)
(b) What is the energy loss in the journey from A to D ?
(c) The total length $ABCD$ is 320 m. Find the average friction f between the track and the coaster.
(d) In the journey from D to E , find the braking force F acting on the coaster.

(a) $PE_{\text{loss}AD} = 30(9.81)(200) = 58860 \text{ J}$

(b) $E_{\text{loss}} = PE_{\text{loss}AD} - KE_{\text{gain}AD} = 58860 - \frac{1}{2}(200)20^2 = 18860 \text{ J}$

(c) $WD_{\text{friction}} = 18860 = 320(f)$
 $f = 58.9 \text{ N}$

(d) $KE_{\text{loss}} = WD_{\text{friction}} \quad f = 2667 \text{ N}$
 $\frac{1}{2}(200)20^2 = 15f$



8. Two blocks are hung over a smooth pulley and the 3 kg block move downward 1 m and reach the floor while the 2 kg block move upward 1 m.

- (a) What would be the P.E. lost by the 3 kg block? (b) What would be the P.E. gain by the 2 kg block?

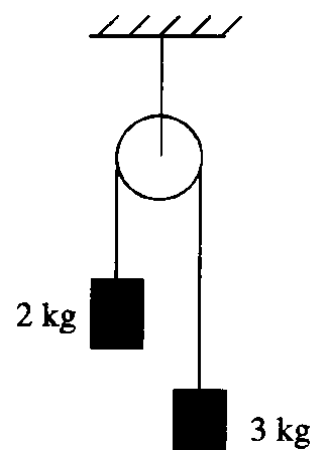
$$PE_{\text{loss}} = 3(9.81)(1) = 29.43J$$

$$PE_{\text{gain}} = 2(9.81)(1) = 19.62J$$

- (c) What would be the velocity of the 2 kg block?

$$PE_{\text{loss}} = PE_{\text{gain}} + KE_{\text{gain}}$$

$$29.52 = 19.62 + \frac{1}{2}(5)v^2 \quad v = 2ms^{-1}$$

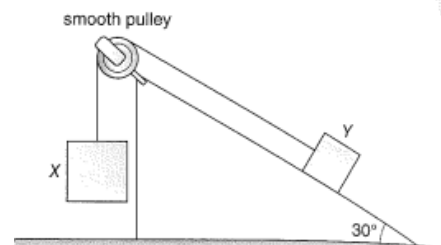


9. Two objects X and Y are connected by a light inextensible string over a smooth pulley as shown. The masses of X and Y are 0.2 kg and 0.1 kg respectively. Y is placed on a rough plane which is inclined at 30° to the horizontal. The system is released from rest.

- (a) When X has fallen a distance of 0.5 m, it attains a speed of $1 ms^{-1}$

- (i) Find the potential energy lost by X
(ii) Find the kinetic energy gained by Y
(iii) Find the average friction acting on Y.

- (b) Describe briefly what happens to Y after X has reached the ground



- (a) (i) PE lost by X = mgh
 $= 0.2 \times 9.81 \times 0.5$
 $= 0.981 J$

1M

- (ii) The speed of X is equal to that of Y.

$$KE_{\text{gained by Y}} = \frac{1}{2}mv^2$$

$$= \frac{1}{2} \times 0.1 \times 1^2$$

$$= 0.05 J$$

1M

- (iii) By the law of conservation of energy,

$$PE_{\text{lost by X}} = KE_{\text{gained by X and Y}} + PE_{\text{gained by Y}} + \text{work done against friction}$$

1M

$$0.981 = \frac{1}{2} \times (0.1 + 0.2) \times 1^2 + 0.1 \times 9.81 \times 0.5 \sin 30^\circ + f \times 0.5$$

$$f = 1.17 N$$

1A

$$\therefore \text{The average friction acting on Y is } 1.17 N.$$

- (b) It moves a little further upwards and then stops.

1A

STRATEGY

Since the two masses are connected by an inextensible string, they move at the same speed.

STRATEGY

Identify what kinds of energy exist in the system and how the energy changes. If friction presents, there must be some energy lost in overcoming the friction.

STRATEGY

Distance travelled by Y is the same as that by X.

STRATEGY

It is moving with a certain velocity when the accelerating force disappears. Therefore, it continues to move a little further upwards. It does not slide downwards as the weight component down the plane is smaller than the maximum friction.

10. A pendulum bob is released from rest from a point P 0.8 m above its lowest position Q , as shown in the above diagram. Neglecting air resistance, find the speed of the bob when it reaches Q

A. 2.8 ms^{-1}

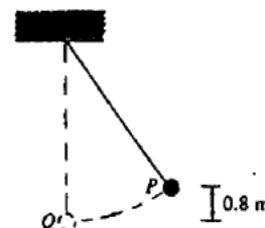
B. 4 ms^{-1}

C. 8 ms^{-1}

D. It cannot be determined since the mass of the bob is not given

$$PE_{\text{loss}} = KE_{\text{gain}} \quad v = 3.96 \text{ ms}^{-1}$$

$$(0.8)(9.81)(m) = \frac{1}{2}(m)v^2$$



B

11. A trolley runs down a friction-compensated runway as shown above. Which of the following statements is true?

A. The kinetic energy of the trolley increases with time.

B. The potential energy of the trolley increases with time.

C. The frictional force acting on the trolley is zero.

D. The resultant force acting on the trolley is zero.



D

12. Which statement below about a raindrop falling at a constant terminal velocity is correct?

A. No work is done on the raindrop by the gravitational force.

B. As the raindrop falls, all its gravitational potential energy loss is converted into kinetic energy gain.

C. The only force acting on the raindrop is its weight.

D. No net force is acting on the raindrop.

A * As the raindrop falls, its gravitational potential energy decreases. Hence, work is done on the raindrop by gravitational force.
B * Since the raindrop is falling at a constant velocity, its kinetic energy remains unchanged.
Note: As the raindrop falls, all its gravitational potential energy loss is converted into work done against air resistance.
C * Apart from its weight, the raindrop is also acted by air resistance.
D ✓ This is because the raindrop is falling at a constant velocity.
Note: The weight of the raindrop and air resistance just cancel out each other.



raindrop falling at a constant velocity

D

13. A ball moves along a smooth curved rail and passed P at a certain speed as shown in the figure above. Neglecting air resistance and friction, which of the following statements about the motion of the ball are correct?

(1) The ball has maximum kinetic energy at R

(2) The speed of the ball at S is the same as that at Q

(3) The ball can never reach T

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

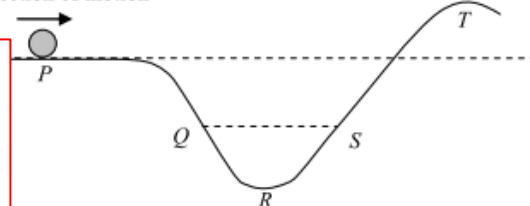
Since the rail is smooth, there is no work done against friction. Hence the mechanical energy of the ball conserves.

(1) ✓ At R , the ball reaches the lowest position. Hence it has minimum potential energy and maximum kinetic energy.

(2) ✓ At S and Q , the ball is at the same height. Hence it has the same potential energy and thus the same kinetic energy. Therefore the speed of the ball at S is the same as that at Q .

(3) * At P , the ball has non-zero kinetic energy. Hence it may reach a higher position by converting its kinetic energy to potential energy.

direction of motion



A

14. Two identical blocks A and B travel down two different tracks as shown.

Assuming that friction is negligible. Which of the following statements are correct?

(1) The potential energy lost by block A is the same as that of block B when they reach the end of the track.

(2) The velocity of block A is the same as that of block B when they reach the end of the track.

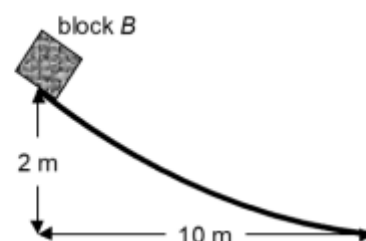
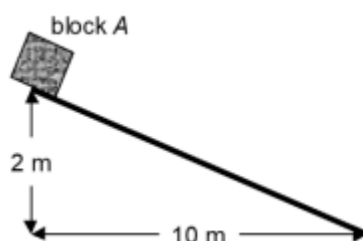
(3) Block A and block B reach the end of the track at the same time.

A. (1) only

B. (1) and (2) only

C. (2) and (3) only

D. (1), (2) and (3)



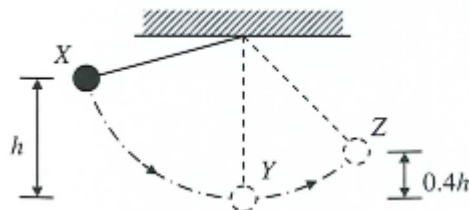
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A

15. A pendulum bob is released from rest at position X which is at a height h above its lowest position Y and then passes through position Z. Position Z is $0.4h$ above Y. Assuming air resistance is negligible, find the ratio of the speed of the bob at position Y to that of position Z

- A. 1.29 : 1
B. 1.58 : 1
C. 1.67 : 1
D. 2.50 : 1

Let v_Y and v_Z be its speed at position Y and position Z respectively.
By conservation of energy,
 $mgh = \frac{1}{2}mv_Y^2$
 $v_Y = \sqrt{2gh}$
and $mg(0.6h) = \frac{1}{2}mv_Z^2$
 $v_Z = \sqrt{1.2gh}$
 $\therefore \frac{v_Y}{v_Z} = \frac{\sqrt{2gh}}{\sqrt{1.2gh}} = 1.29 : 1$



A

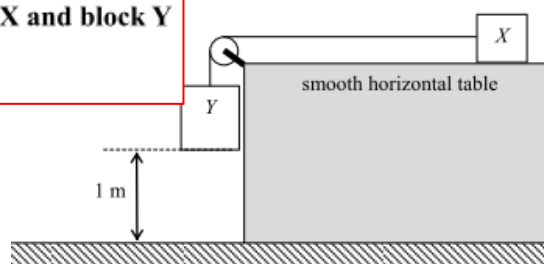
16. 2017 DSE Q.10

Blocks X and Y are connected by a light inextensible string passing over a fixed frictionless light pulley as shown. The mass of X and Y are 0.5 kg and 1 kg respectively. Initially, Y is 1 m above the ground and the string is taut. The system is then released from rest. What is the speed of Y just before it reaches the ground?

- A. 3.62 m s^{-1}
B. 4.43 m s^{-1}
C. 6.26 m s^{-1}
D. 9.81 m s^{-1}

Loss in G.P.E. of block Y = Gain in K.E. of block X and block Y

$$(1)(9.81)(1) = \left(\frac{1}{2}\right)(1 + 0.5)(v^2) \Rightarrow v = 3.62 \text{ m s}^{-1}$$



A

17. Two blocks of respective masses 2 kg and 5 kg are connected by a light inextensible string which passes over a smooth fixed light pulley as shown. The system is released from rest when the 5-kg block is 3 m above the ground. What is the speed of the 5-kg block just when reaching the ground?

Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

- A. 5.0 m s^{-1}
B. 6.0 m s^{-1}
C. 6.5 m s^{-1}
D. 7.7 m s^{-1}

Method 2

Let v be the speed of the 5-kg block just when reaching the ground.

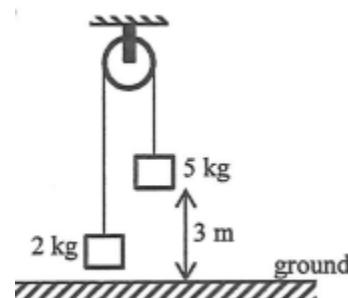
Note that the speeds of the two blocks are equal.

By the law of conservation of mechanical energy,

Gain in total kinetic energy of the two blocks = Loss in total gravitational potential energy of the two blocks

$$\frac{1}{2}(2)v^2 + \frac{1}{2}(5)v^2 = (2)(9.81)(-3) + (5)(9.81)(3)$$

$$v = 5.0 \text{ m s}^{-1}$$



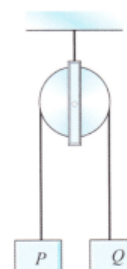
A

18. Two blocks, P and Q, are connected by a light string passing over a smooth pulley as shown above. The mass of P is greater than that of Q. The two blocks are released from rest. Which of the following statements is/are correct?

- (1) The potential energy lost by P is equal to the potential energy by Q
(2) The kinetic energy gained by P is greater than the kinetic energy gained by Q
(3) The potential energy lost by P is equal to the sum of kinetic energies gained by P and Q

- A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

- (1) is incorrect. As P falls, the loss in potential energy of P becomes the gain in potential energy of Q and the gain in kinetic energy of P and Q.
 $\therefore PE_{\text{lost by P}} = PE_{\text{gained by Q}} + KE_P + KE_Q$
(2) is correct. The speed of P and Q are the same. Their kinetic energy are
 $KE_P = \frac{1}{2}m_P v^2$,
 $KE_Q = \frac{1}{2}m_Q v^2$.
Since $m_P > m_Q$, $KE_P > KE_Q$.
(3) is incorrect. The potential energy lost by P is greater than the sum of kinetic energies gained by P and Q. See (1) above.



A

19. A car stopped after emergency braking. The skid mark left by the car was 22.3 m long. Assume that the friction between the road and tyres was 0.65 times that of the weight of the car. Estimate the speed of the car when it began to skid.

- A. 5.38 ms^{-1}
B. 12.0 ms^{-1}
C. 16.2 ms^{-1}
D. 16.9 ms^{-1}

Assume the deceleration during braking is only contributed by the friction force between the road and the tyre, then the kinetic energy lost by the car is equal to the work done against friction. Then we have

$$\begin{aligned}\frac{1}{2}mv^2 - \frac{1}{2}mu^2 &= Fs \\ \Rightarrow \frac{1}{2}m(0)^2 - \frac{1}{2}mu^2 &= (-0.65mg)(22.3) \\ \Rightarrow u^2 &= 2 \times (0.65)(10)(22.3) \\ &= 289.9 \\ \Rightarrow u &= 17.0 \text{ m s}^{-1}\end{aligned}$$

D

20. 1995/II/9

A bullet of mass 0.02 kg travelling at 450 ms^{-1} enters a wooden block as shown in Figure (a) and leaves it with a speed 400 ms^{-1} . In Figure (b), what is the maximum of identical blocks that the mass bullet can pass through?

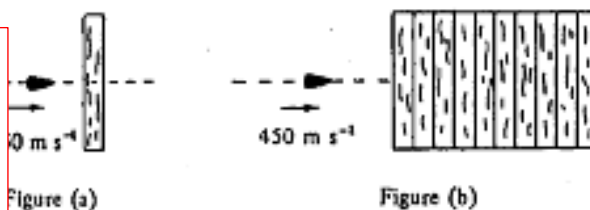
- A. 3
B. 4
C. 5
D. 9

$$KE_{\text{loss per block}} = \frac{1}{2}(0.02)450^2 - \frac{1}{2}(0.02)400^2$$

$$= 425 \text{ J}$$

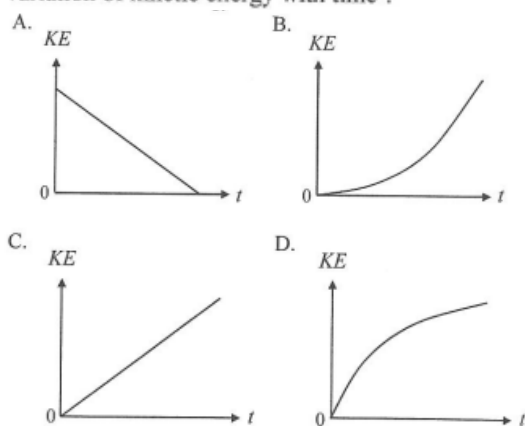
$$425n = \frac{1}{2}(0.02)450^2$$

$$n = 4.76$$



B

21. An object slides down a smooth plane from rest as shown in the figure above. Which of the following graphs best shows the variation of kinetic energy with time?



As it is a graph of energy against time, it must be a curve.

At $t = 0$, the object starts from rest, thus $KE = 0$.

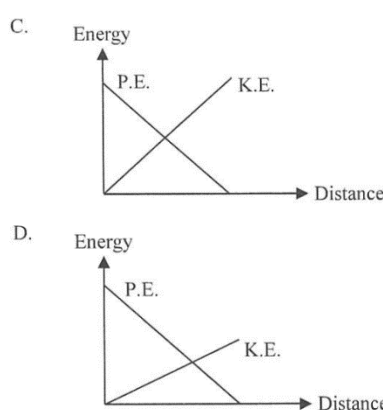
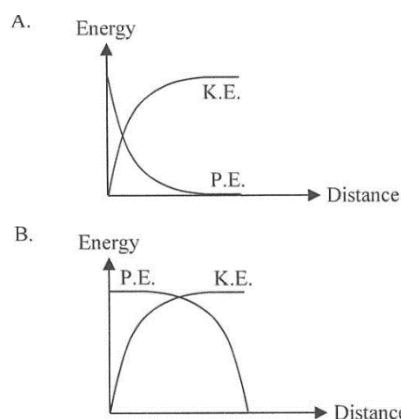
As time increases, the object moves down with loss of PE , thus KE increases.

As $KE \propto v^2$, the curve is a quadratic curve opening upwards.

B

22. 2007/II/4

A skier slide down a slope as shown in the diagram below. Assume constant friction along the slope, which of the following graphs best describes the change of energy of skier with distance down the slope?

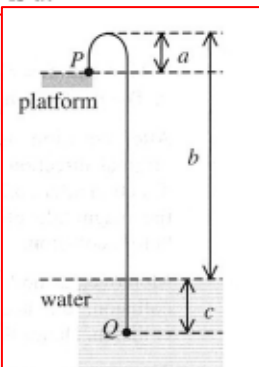
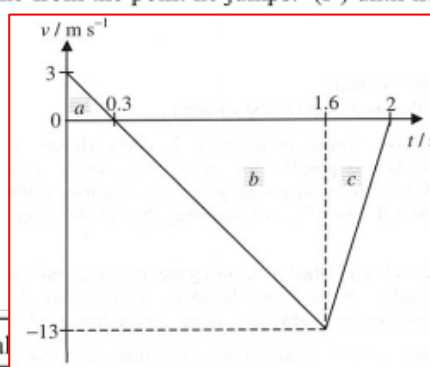
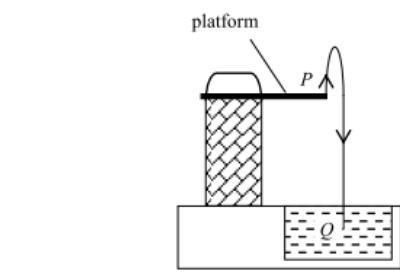


D

警告:

必究

23. A diver jumps up vertically in the air from a high platform and falls into water. The $v-t$ graph below shows the variation of the velocity of the diver against time from the point he jumps. (P) until he is at lowest point (Q) in the water.



C

Which of the following is correct?

not to scale

	Total distance travelled from P to Q (m)	Height of the platform above water surface (m)
A.	8.9	8
B.	10.6	10
C.	11.5	8
D.	11.5	10.6

At $t = 0.3$ s, the diver is momentarily at rest, so he/she reaches highest point.
At $t = 1.6$ s, the slope of the $v-t$ graph, i.e. the acceleration of the diver, changes sharply. This indicates the unbalanced force acting on him/her changes sharply. So he/she reaches the water surface.

Power

- Power is the rate at which energy is transferred or work is done.

$$\text{Power} = \frac{\text{Work done}}{\text{time}} \text{ or } \frac{\text{Energy transferred}}{\text{time}}$$

$$\text{Unit: Watt (W)} = \text{J s}^{-1}$$

- For an object moving at a **constant velocity**, F is the applied force that opposes the resisting force acting on the object.

$$\text{Power} = Fv$$

24. An athlete of mass 48 kg runs up a flight of steps in 60 s. If the vertical height of the top of the steps above ground is 50 m, find the average power of the athlete.

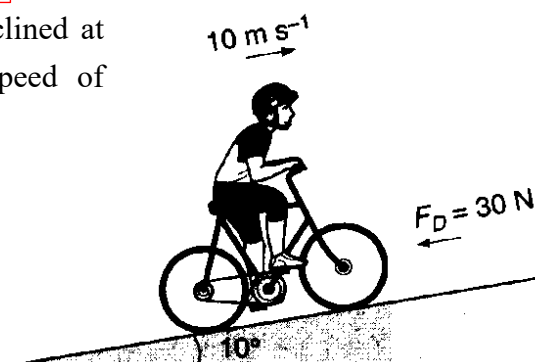
$$PE_{\text{gain}} = (48)(9.81)(50) = P(60)$$

$$P = 392.4 \text{ W}$$

25. A 50 kg cyclist rides a bicycle of mass 30 kg up a road inclined at 10° to the horizontal. The cyclist maintains a constant speed of 10 ms^{-1} uphill. The frictional force is 30 N downhill.

- (a) What is the increase in potential energy of the cyclist and bicycle in 10 s?

$$PE_{\text{gain}} = 10(10) \sin 10^\circ (9.81)(80) = 13628 \text{ J}$$



- (b) What is the work done and the power of the cyclist in 10 s?

$$WD = 10(10)(30) + 13628 = 16628 \text{ J}$$

$$P(10) = 15528 \text{ J}$$

$$P = 1552.8 \text{ W}$$

26. A car traveling at a constant velocity of 30 ms^{-1} on a level road. What is the power output of the car engine if the total of the air resistance and friction acting on the car is 800 N .

$$WD = 30(800) = 24000 \text{ J}$$

$$P = 24000 \text{ W}$$

27. David is pushing a load of mass 15 kg on a horizontal floor with a horizontal force of 50 N . The load moves at a constant speed of 2 m s^{-1} .

- (a) Find the friction acting on the load.
(b) Find the power delivered to overcome the friction.

- (a) Since the load is moving at a constant speed, $\downarrow$
the friction acting on the load is -50 N . (1A)
(b) Applying $P = Fv$, the power delivered to overcome friction is
 $P = 50 \times 2$
 $= 100 \text{ W}$ (1M+1A)

28. 1996/II/7

A motor car of masses 2500 kg accelerates from rest to a speed of 20 ms^{-1} in 5 s on a level road. Find the average power of the car.

- A. 10 kW
B. 50 kW
C. 100 kW
D. 200 kW

Method 1
 $E = Fv$
 $P = \frac{E}{t} = \frac{Fv}{t} = \frac{(2500)(10)(20)}{5} = 100000 \text{ W} = 100 \text{ kW}$

Method 2
 $P = \frac{E}{t} = \frac{\frac{1}{2}mv^2}{t} = \frac{\frac{1}{2}(2500)(20)^2}{5} = 100000 \text{ W} = 100 \text{ kW}$

C

29. 1997/II/8

A machine lifts up a load of 1200 N to a height of 1.5 m in 2 s . Find the average output power of the machine.

- A. 400 W
B. 900 W
C. 1800 W
D. 3600 W

Total energy needed = increase in P.E. = $(1200)(1.5) = 1800 \text{ J}$
Average power = $\frac{\text{Energy}}{\text{Time}} = \frac{1800}{2} = 900 \text{ W}$

B

30. 2012 DSE MC Q.9

An object of mass 0.5 kg is raised vertically from the ground by a motor. The object is raised 2.5 m in 1.5 s with uniform speed. Estimate the output power of the motor. Neglect air resistance. ($g = 9.81 \text{ ms}^{-2}$)

- A. 5.5 W
B. 8.2 W
C. 11 W
D. 16.4 W

The potential energy gained by the object = $(0.5)(9.81)(2.5) = 12.2625 \text{ J}$
The output power of the motor = $\frac{12.2625}{1.5} = 8.175 \text{ W}$

B

31. 1999/II/7

A motor car of mass 2000 kg accelerated from rest at 3 ms^{-2} for 4 s on a straight road. Find the average useful output power of the engine of the car.

- A. 24 kW
B. 36 kW
C. 72 kW
D. 144 kW

$$\begin{aligned} a &= 3 \text{ ms}^{-2}, t = 4 \text{ s}, u = 0 \text{ ms}^{-1} \\ s &= ut + \frac{1}{2}at^2 = \frac{1}{2}(3)(4)^2 = 24 \text{ m} \\ F &= ma = 2000 \times 3 = 6000 \text{ N} \\ W &= Fs = 6000 \times 24 = 144000 \text{ J} \\ P &= \frac{W}{t} = \frac{144000}{4} = 36000 \text{ W} = 36 \text{ kW} \end{aligned}$$

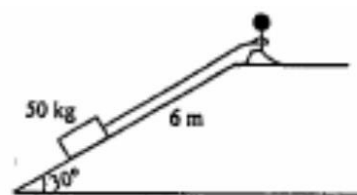
B

32. 1998/II/10

A man pulls a 50 kg block up a smooth inclined plane making an angle 30° with the horizontal as shown above. The inclined plane is 6 m long and the block is pulled from the bottom of the inclined plane to the top in 30 s. Find the average useful output power of the man.

- A. 5 W
B. 10 W
C. 50 W
D. 87 W

$$\begin{aligned} \text{Increase in PE} &= mgh \\ &= 50(10)(6 \sin 30^\circ) \\ &= 1500 \text{ J} \\ \text{Average useful output power} &= \frac{1500}{30} = 50 \text{ W} \end{aligned}$$



C

33. 2000/II/11

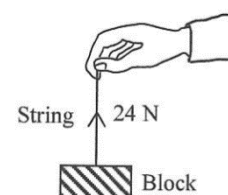
A 2 kg block is initially at rest on the ground. A machine is used to pull up the block as shown above. If the tension in the string is kept at 24 N and the block reaches a height of 4 m in 2 s, which of the following statements is/are correct?

- (1) The potential energy of the block is increasing when the block is rising
(2) The kinetic energy of the block is increasing when the block is rising
(3) The average power developed by the machine during the two second is 40W

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

$$(1) \text{ P.E.} = mgh \quad h \uparrow \Rightarrow \text{P.E.} \uparrow$$

$$\begin{aligned} (2) \text{ By } F &= mg + ma \\ 24 - (2)(10) &= 2a \quad \therefore a = 2 \text{ ms}^{-2} > 0 \\ \therefore \text{The block is accelerating.} \\ \therefore \text{Velocity and hence K.E. is increasing.} \\ (3) \text{ average power} &= \frac{W}{t} = \frac{F \cdot s}{t} = \frac{(24)(4)}{2} = 48 \text{ W} \end{aligned}$$



C

34. 2017 DSE Q.11

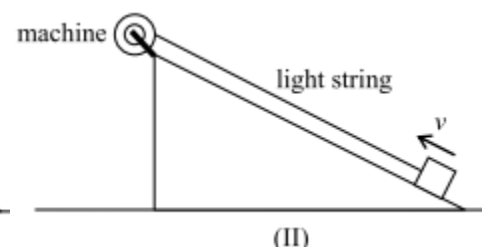
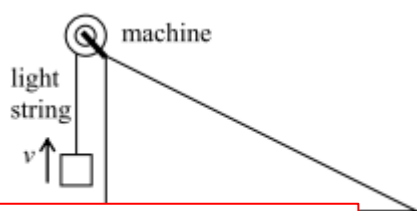
A machine is fixed at the top of a smooth inclined plane. Two methods, (I) and (II), are used to lift a block from the ground to the top of the inclined plane by the machine.

- (I) Pull the block vertically upward at a uniform speed v .
(II) Pull the block up along the inclined plane at the same uniform speed v .

Which of the following statements correctly compare(s) the two methods?

- (1) The tension in the string is the same.
(2) The average output power of the machine is the same.
(3) The work done by the machine on the block is the same.

- A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only



Energy gain by the block in (I) and (II) are the same.
different distance, same speed $\Rightarrow$ different time $\Rightarrow$ different power ($P = E/t$)
different distance $\Rightarrow$ different force

B